



Assignment of bachelor's thesis

Title:	Downscaling climate data with physically-constrained single-image super resolution
Student:	Tomáš Becza
Supervisor:	Ing. Ondřej Podsztavek, Ph.D.
Study program:	Informatics
Branch / specialization:	Artificial Intelligence 2021
Department:	Department of Applied Mathematics
Validity:	until the end of summer semester 2026/2027

Instructions

Climate projections (i.e. simulations of climate typically until 2100) with high spatial resolution are necessary for local decision-making, such as urban planning, agriculture, or forest management. However, only climate projections with low spatial resolution are usually available, which are often insufficient. Therefore, we must increase their spatial resolution (i.e. downscale them) to 1 km and higher. This task can be approached with single-image super-resolution models. However, these models often do not account for the nature of climate data. Incorporating physical constraints can improve their performance. The goal is to assess the downscaling of climate data with physically-constrained single-image super-resolution models.

1. Review literature on downscaling with physically-constrained single-image super-resolution models.
2. Implement a baseline (e.g. bilinear interpolation), single-image super-resolution model (e.g. an enhanced deep residual network) and at least two physical constraints.
3. Experiment on the observational gridded data set from Regionales Klimainformationssystem für Sachsen, Sachsen-Anhalt und Thüringen (ReKIS) and a climate projection from the European Coordinated Regional Downscaling Experiment (EURO-CORDEX).
4. Evaluate the baseline and models (without and with physical constraints) with relevant metrics (e.g. root-mean-square error).
5. Discuss results in the context of the literature review.



Bibliography

Harder, Paula, Alex Hernandez-Garcia, Venkatesh Ramesh, et al. 2023. 'Hard-Constrained Deep Learning for Climate Downscaling'. Journal of Machine Learning Research 24 (365): 1–40.

Košťál, Petr, Pavel Kordík, and Ondřej Podsztavek. 2025. 'Downscaling Climate Projections to 1 Km with Single-Image Super Resolution'. Paper presented at NeurIPS 2025 Workshop on Tackling Climate Change with Machine Learning. Climate Change AI, December 7. <https://www.climatechange.ai/papers/neurips2025/86>.

Lim, Bee, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. 2017. 'Enhanced Deep Residual Networks for Single Image Super-Resolution'. 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), July, 1132–40. <https://doi.org/10.1109/CVPRW.2017.151>.